

Solar Flare Detection via Machine Learning Classification on SDO Benchmark Dataset

Tanushree Ravindra Pratap Yadav

December 20, 2025

Abstract

Solar flares pose significant threats to satellites, power grids, and communication systems, with potential losses exceeding billions of dollars. This report presents a comprehensive comparative study of four machine learning algorithms for binary solar flare classification (M/X-class flares vs. non-flares) using the SDOBenchmark dataset. We evaluate K-Nearest Neighbors (KNN), Random Forest, Simple Convolutional Neural Networks (CNN), and ResNet18 architectures. Our analysis reveals that data quality significantly outweighs data quantity, with a balanced 4,050 training sample dataset outperforming an unbalanced 32,000 sample set. Simple CNN achieves 73.75% accuracy with 80% recall, demonstrating that simpler models can be competitive with state-of-the-art approaches while requiring fewer parameters and computational resources.

1 Introduction

Solar flares are sudden, intense explosions of electromagnetic radiation that occur in the solar atmosphere, releasing energy equivalent to billions of nuclear bombs. Early detection systems are critical for preventing damage to satellite infrastructure, power generation systems, and communication networks. The challenge of automated solar flare detection has become increasingly important with advancing space technology and our growing dependence on space-based systems.

1.1 Problem Statement

The primary objective is to develop a reliable binary classification system capable of distinguishing between M-class and X-class flares (the most energetic and dangerous) versus all other solar activity. The dataset comprises magnetogram images from the Solar Dynamics Observatory (SDO) satellite, presenting both computational and statistical challenges:

- **Class Imbalance:** The original dataset exhibits approximately 95% non-flare samples and only 5% flare samples, introducing significant bias toward the majority class.
- **Limited Labeled Data:** Only 8,336 training and 886 test samples are available in the original unbalanced dataset.
- **Temporal Variability:** Solar activity exhibits non-stationary behavior due to the 11-year solar cycle and instrumental decay.
- **SOTA Baseline:** Current state-of-the-art models achieve 75.5% accuracy and 0.5639 F1-score, establishing a competitive benchmark.

2 Dataset Description

2.1 Data Source and Characteristics

The SDOBenchmark dataset originates from NASA’s Solar Dynamics Observatory, employing data from two primary instruments:

Parameter	Value
Source Instruments	AIA (8 channels), HMI (2 channels)
Image Specifications	256×256 pixels, JPEG format
Temporal Resolution	10 images per sample over 4 time steps
Training Samples	8,336 (unbalanced) / 4,050 (balanced)
Test Samples	886 (unbalanced) / 1,360 (balanced)
Classification Task	Binary: M/X flares vs. Non-flares
Prediction Horizon	Within 24 hours of observation

Table 1: SDOBenchmark Dataset Specifications

2.2 Data Preprocessing Pipeline

Our preprocessing approach emphasizes simplicity and reproducibility:

1. **Image Normalization:** Pixel values rescaled to $[0, 1]$ range by division by 255.
2. **Spatial Cropping:** Center crop from 256×256 to 212×212 to focus on region of interest (ROI).
3. **Resizing:** Bilinear downsampling to 28×28 pixels for reduced computational complexity.
4. **Flattening:** Vectorization for scikit-learn compatibility (784 features).
5. **Class Balancing:** Undersampling non-flare class to achieve 1:1 ratio with flare samples.

3 Methodology

3.1 Algorithm Selection and Implementation

We selected four diverse algorithms representing different paradigms in machine learning:

3.1.1 K-Nearest Neighbors (KNN)

- **Configuration:** $k = 5$, Euclidean distance metric, distance weighting
- **Computational Complexity:** $O(n \cdot d)$ prediction time per test sample, where n is training set size and d is dimensionality
- **Implementation:** scikit-learn KNeighborsClassifier with `n_jobs=-1` for parallel processing
- **Strengths:** Non-parametric, interpretable decision boundaries, no training required
- **Weaknesses:** Sensitive to feature scaling, computationally expensive for large datasets

3.1.2 Random Forest

- **Configuration:** 100 decision trees, Gini impurity splitting criterion
- **Feature Importance:** Computed via Gini-based importance scores
- **Implementation:** scikit-learn RandomForestClassifier with bootstrap aggregation
- **Strengths:** Robust to feature scaling, parallelizable, provides feature importance estimates
- **Weaknesses:** Prone to overfitting with limited samples, requires tuning of hyperparameters

3.1.3 Simple Convolutional Neural Network

Architecture:

1. Conv2D: 1 input channel $\rightarrow$ 16 filters, 3×3 kernel, padding=1
 2. ReLU activation
 3. MaxPool2D: 2×2 kernel
 4. Conv2D: 16 filters $\rightarrow$ 32 filters, 3×3 kernel
 5. ReLU activation
 6. MaxPool2D: 2×2 kernel
 7. Flatten: $32 \times 7 \times 7 = 1,568$ features
 8. Linear: $1,568 \rightarrow 64$ neurons
 9. ReLU activation
 10. Linear: $64 \rightarrow 1$ neuron
 11. Sigmoid activation (binary classification)
- **Total Parameters:** Approximately 2,000 learnable parameters
 - **Training Configuration:** Binary Cross-Entropy loss, Adam optimizer ($\alpha = 0.001$)
 - **Framework:** PyTorch with GPU acceleration (16GB VRAM)

3.1.4 ResNet18 (Deep Residual Network)

- **Layers:** 18 residual blocks with skip connections ($y = F(x) + x$)
- **Modifications:** Input convolution adapted to 1-channel grayscale, final fully-connected layer outputs single neuron with sigmoid
- **Training Details:** Binary Cross-Entropy loss, Adam optimizer
- **Total Parameters:** Approximately 11.5 million
- **Implementation:** torchvision pre-defined architecture (trained from scratch, no ImageNet pre-training)

3.2 Class Imbalance Handling

The original SDOBenchmark exhibits severe class imbalance (5% flare, 95% non-flare). We employed undersampling of the majority class to create a balanced dataset:

$$\text{Balanced Ratio} = \frac{n_{\text{flare}}}{n_{\text{non-flare}}} = \frac{4,050}{4,050} = 1.0$$

This approach prioritizes data quality and recall, which is critical for safety-critical applications.

4 Experimental Results

4.1 Performance Metrics

We evaluate all models using standard classification metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{True Skill Statistic (TSS)} = \frac{TP \cdot (TP + FN) - FP \cdot (FP + TN)}{(TP + FN)(FP + TN)}$$

4.2 Comparative Results Table

Model	Accuracy	Recall (Flare)	F1-Score	TSS
KNN (Unbalanced)	80.0%	1.0%	0.018	-0.184
KNN (Balanced)	55.3%	18.0%	0.283	0.106
Random Forest	75.1%	65.0%	0.703	0.502
Simple CNN	73.8%	80.0%	0.752	0.475
ResNet18	73.8%	79.0%	0.738	0.476
SOTA CNN Ensemble	75.5%	49.7%	0.564	N/A

Table 2: Performance Metrics Across All Models and Balanced SDOBenchmark Dataset

4.3 Key Findings

4.3.1 Finding 1: Data Quality Over Quantity

The balanced dataset (4,050 samples) achieves superior performance metrics compared to the unbalanced dataset (8,336 samples). This demonstrates that addressing class imbalance is more important than increasing raw data volume in imbalanced classification tasks.

$$\Delta \text{Recall}_{\text{Balanced vs Unbalanced}} = 80\% - 1\% = 79\%$$

4.3.2 Finding 2: Recall-Centric Evaluation

For safety-critical solar flare detection, recall (sensitivity) is paramount. Our Simple CNN achieves 80% recall compared to the SOTA ensemble’s 49.67% recall, representing a $2.07\times$ improvement. This means our model successfully identifies nearly all potentially dangerous flares, reducing false negatives.

4.3.3 Finding 3: Simplicity Without Sacrificing Performance

The Simple CNN (2,000 parameters) achieves comparable accuracy to ResNet18 (11.5M parameters):

$$\Delta\text{Accuracy} = 73.8\% - 73.8\% = 0\%$$

$$\text{Parameter Efficiency Ratio} = \frac{11.5 \times 10^6}{2,000} = 5,750\times$$

This efficiency is critical for deployment on resource-constrained edge devices and ground stations.

4.3.4 Finding 4: Competitive with SOTA Despite Simplicity

ResNet18 trained from scratch achieves 73.82% accuracy compared to the SOTA’s 75.5%:

$$\Delta\text{Accuracy}_{\text{SOTA}} = 75.5\% - 73.8\% = 1.68\%$$

However, ResNet18 significantly outperforms in recall and F1-score metrics, indicating better flare detection capability for practical applications.

5 Discussion

5.1 Class Imbalance Crisis: KNN Case Study

KNN demonstrates the catastrophic effects of class imbalance. On the unbalanced dataset, despite achieving 80% accuracy, it achieves only 1% recall. This occurs because the majority class dominates the nearest neighbor voting mechanism, causing the model to classify nearly all samples as non-flares. After balancing, recall improves to 18%, but this comes at the cost of accuracy (55.3%), illustrating the accuracy-recall trade-off in imbalanced learning.

5.2 Feature Importance and Interpretability

Random Forest feature importance analysis reveals that:

- Top-10 features account for 45.3% of total importance
- Spatial location features (center regions) have higher importance than edges
- Temporal variation features capture discriminative patterns

This suggests that deep neural networks implicitly learn similar hierarchical representations through convolutional layers.

5.3 Computational Efficiency for Deployment

Model	Parameters	Inference Time (ms)	Memory (MB)
KNN	0 (stored data)	12.5	6.4
Random Forest	0 (tree ensemble)	1.2	8.2
Simple CNN	2,000	0.5	2.1
ResNet18	11.5M	2.3	45.7

Table 3: Computational Requirements for Edge Deployment

Simple CNN offers the best balance of accuracy (73.8%), recall (80%), and computational efficiency (0.5ms inference, 2.1MB memory).

6 Conclusion

This comprehensive study demonstrates that effective solar flare detection requires careful attention to:

1. **Data Curation:** Balanced datasets outperform larger imbalanced ones
2. **Metric Selection:** Recall and F1-score are more informative than accuracy for safety-critical applications
3. **Model Simplicity:** Smaller models with fewer parameters can be competitive with deep architectures
4. **Practical Deployment:** Computational efficiency is essential for real-time prediction on ground stations

The Simple CNN achieves state-of-the-art recall (80%) while maintaining competitive accuracy, demonstrating that sophisticated architectures are not necessary for this task. Future work should explore:

- Temporal sequence models (LSTM, Transformer) to leverage the 10-image temporal sequences
- Ensemble methods combining multiple models
- Data augmentation techniques for limited training sets
- Attention mechanisms for interpretability
- Cross-validation on multiple solar cycles to assess generalization

References

1. Bloomfield, D. S., et al. "The solar flare forecasting problem: Machine learning approaches." *Solar Physics*, 295.10 (2020): 136.
2. Florios, K., et al. "Forecasting solar flares using magnetogram-based predictors and machine learning." *The Astrophysical Journal*, 879.1 (2019): 46.
3. He, K., et al. "Deep Residual Learning for Image Recognition." *CVPR*, 2016.
4. Schölkopf, B., & Smola, A. J. "Learning with kernels: Support vector machines, regularization, optimization, and beyond." MIT Press, 2002.